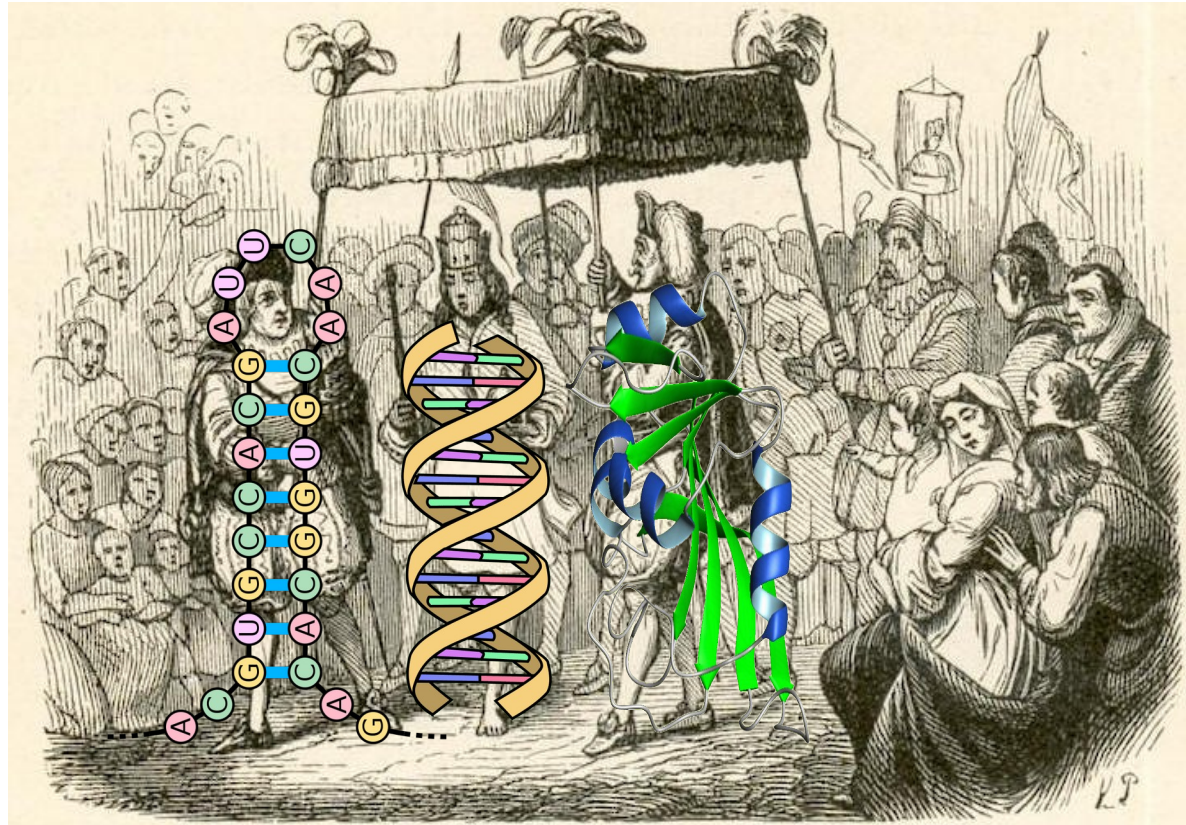


Challenges of Genomics Data Analysis

Nikolay Oskolkov, Group Leader (PI) at NIRI



I remember times when I worked with 10 genetic variants, now we have millions of variants thanks to WGS!

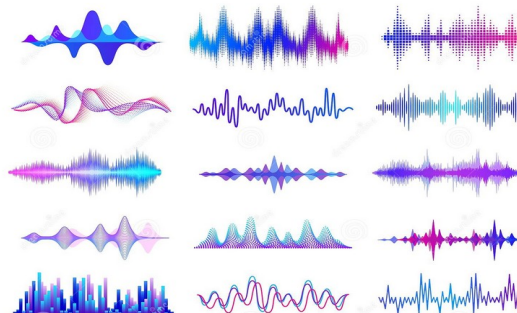


NBS Various types of data around us

Tabular

Google Sheet showing a large table of data with multiple columns and rows.

Sound



Image



Text

Editing Wikipedia articles on Medicine

Engage with editors

Part of the Wikipedia experience is meeting and responding to feedback from other editors. Do not submit your content on the last day. Check back on your article. Read recent comments from the Wikipedia community will likely read and respond to it, and it would be better for you to acknowledge the time they volunteer to public your work. Everything submitted to Wikipedia is reviewed by multiple, not human, editors. You may not get a comment, but if you do, please acknowledge it.

Be accurate

You're editing a resource millions of people use to make critical decisions, so it's really important to be accurate. Wikipedia is used more for medical information than the websites for WebMD, NIH, and the WHO. But with great power comes great responsibility!

Understand the guidelines

Wikipedia editors in the medical area have developed additional guidelines to ensure that the content on Wikipedia is medically sound. Take extra time to read and understand these guidelines. When you edit an article, ensure your changes meet these special requirements. If not, your work is likely to be undone by other editors as they clean up after you. Then take valuable volunteer time away from creating content. If you're not comfortable working under these guidelines, talk to your instructor about an alternative off-wiki assignment.

Watch out for close paraphrasing

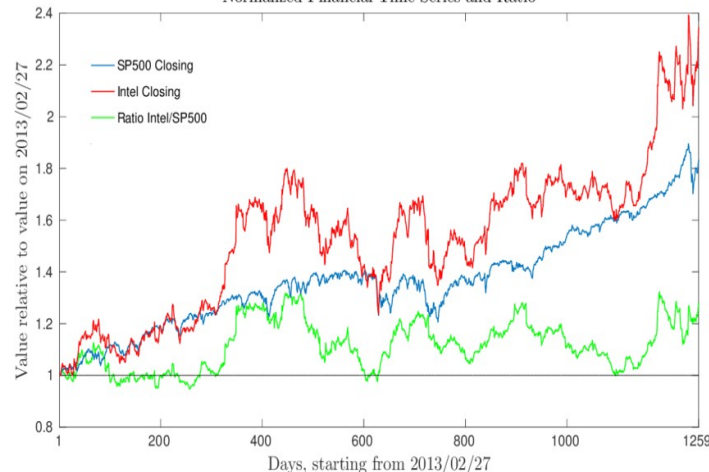
Paraphrasing or close paraphrasing is never okay on Wikipedia and is a violation of your university's academic honor code. It's even worse on Wikipedia, as valuable volunteer time that could be used to create good content is instead used to clean up plagiarized work.

Scared? Don't be!

Everybody on Wikipedia wants to make the best encyclopedia they can. Take the time to understand the rules, and soon you'll be contributing to a valuable resource you use on a daily basis!

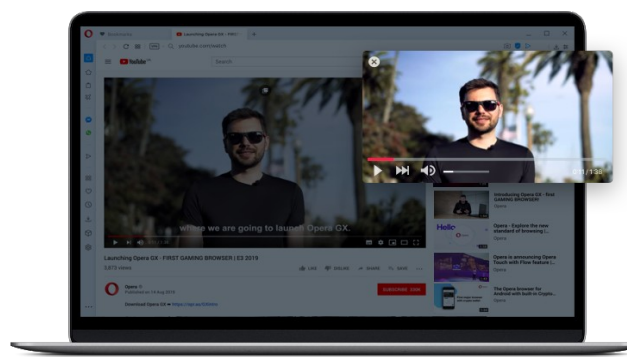
Time Series

Normalized Financial Time Series and Ratio

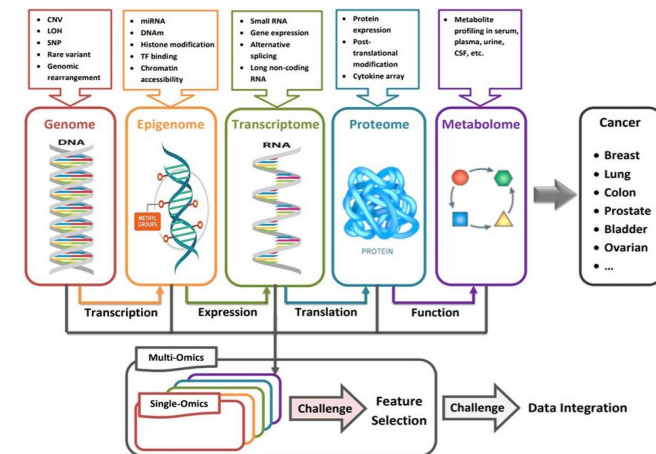


DATA

Video



Omics Data



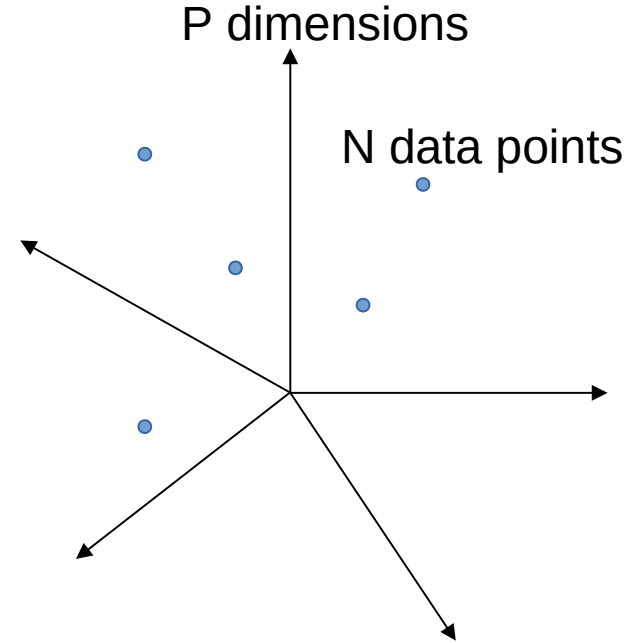
Statistical observations:
e.g. samples, cells etc.

Features: genes, proteins,
microbes, metabolites etc.

N

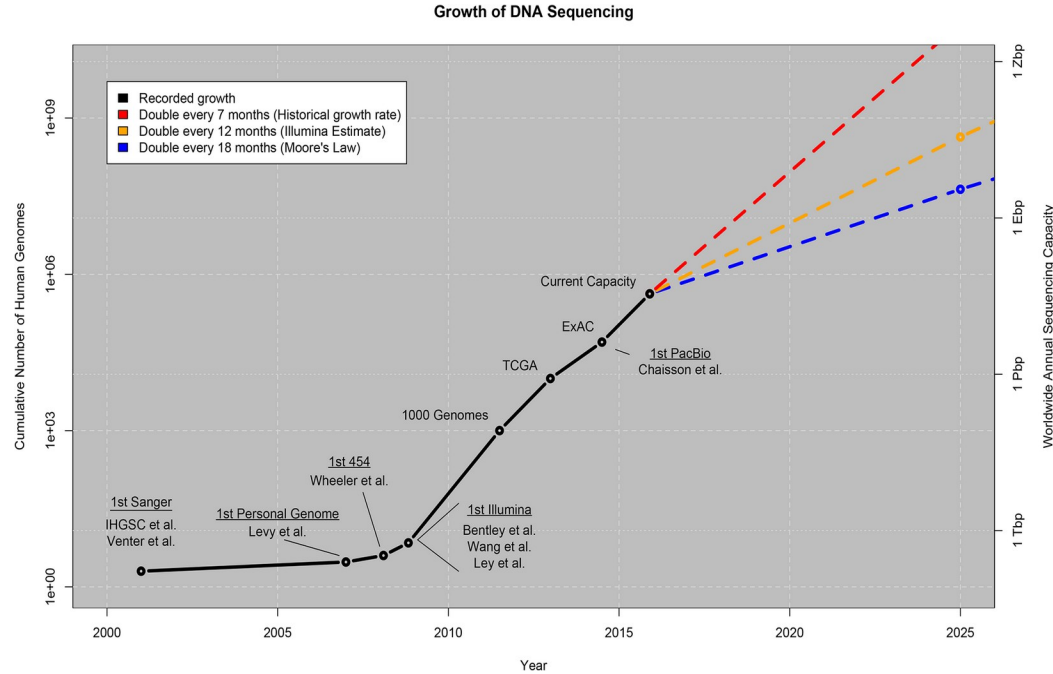
0	3	1	0	2	3	8	1	1	3
1	1	0	0	7	1	2	2	3	3
1	2	2	0	0	6	7	1	2	2
1	2	3	10	0	4	6	1	0	5
3	2	2	1	4	3	2	1	6	0
7	4	4	5	3	9	6	1	6	1
7	1	1	5	2	8	9	1	3	6
5	0	1	6	2	0	0	0	1	5
1	6	3	3	4	6	2	0	1	1
1	2	2	4	1	1	3	0	8	2

P



High Dimensional Data:
 $P \gg N$

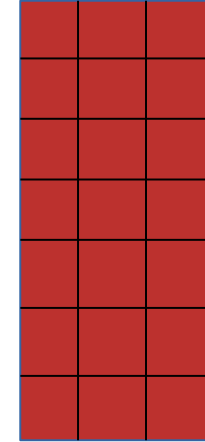
For a robust statistical analysis, one should properly “sample” the P-dimensional space, hence large sample size is required, $N \gg P$



Stephens et al., (2015). Big Data: Astronomical or Genomical? PLoS Biology 13(7)

Genomics / WGS: **Little Data**

$$N_1 \sim 10^3$$

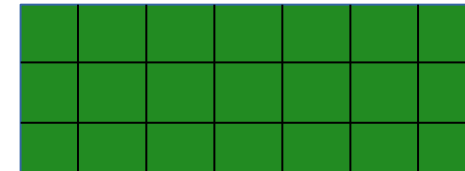


$$P_1 \sim 10^6$$

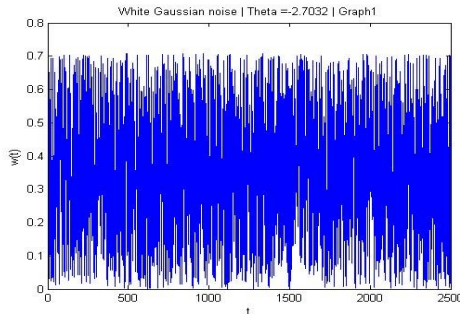
$$N_1 * P_1 = N_2 * P_2 = 10^9$$

scRNAseq: **Big Data**

$$N_2 \sim 10^6$$



$$P_2 \sim 10^3$$

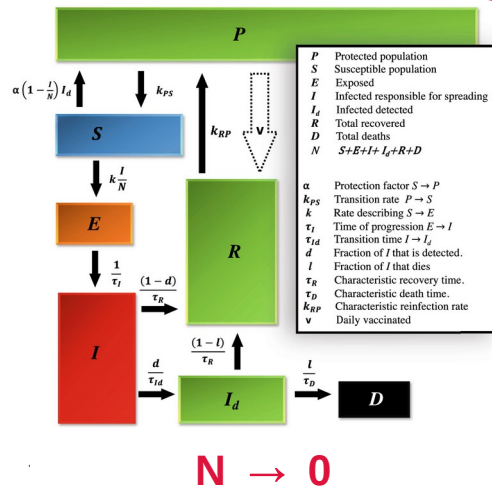


A file with **White Noise** can also take a lot of disk space

P is the number of features (genes, proteins, genetic variants etc.)
 N is the number of observations (samples, cells, nucleotides etc.)

Biology / Biomedicine

Mathematical modeling



Bayesianism



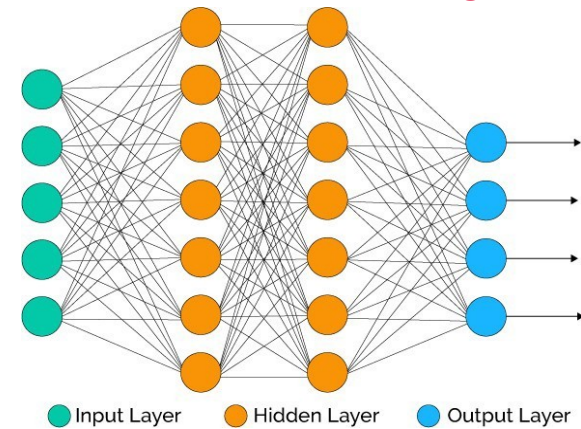
$N \ll P$

Frequentism



$N \approx P$

Machine Learning



$N \gg P$

Hypothesis-driven

Amount of Data

Data-driven

The Curse of Dimensionality

Ex.1

$$Y = \alpha + \beta X$$

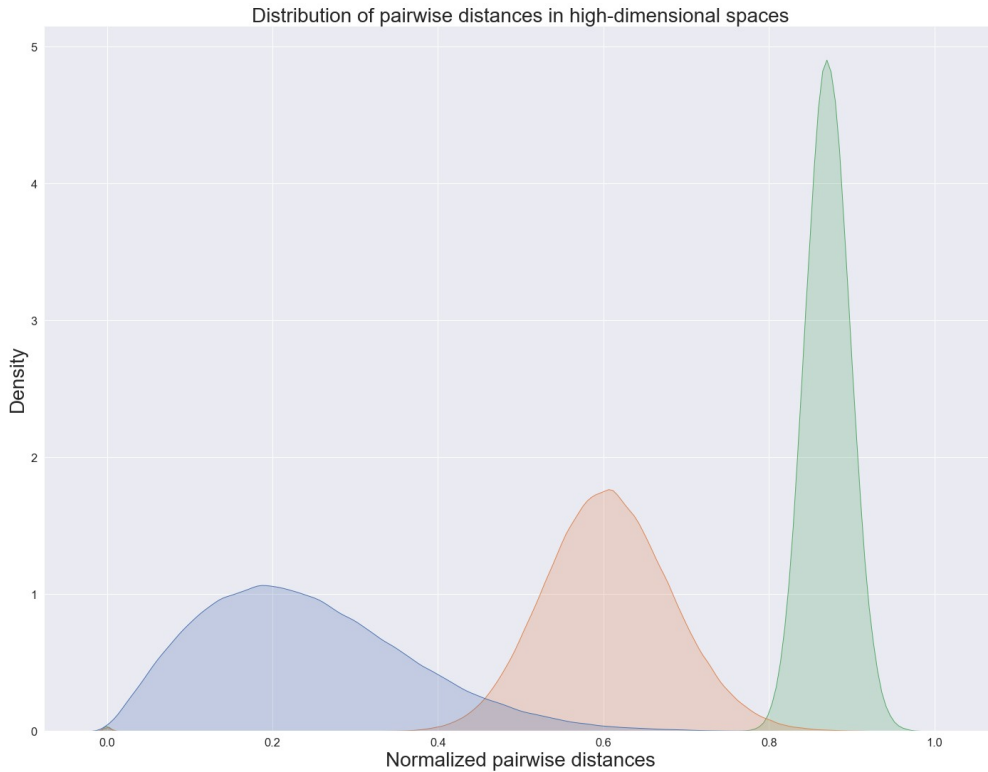
$$\beta = (X^T X)^{-1} X^T Y$$

$$(X^T X)^{-1} \sim \frac{1}{\det(X^T X)} \dots \rightarrow \infty, \quad n \ll p$$

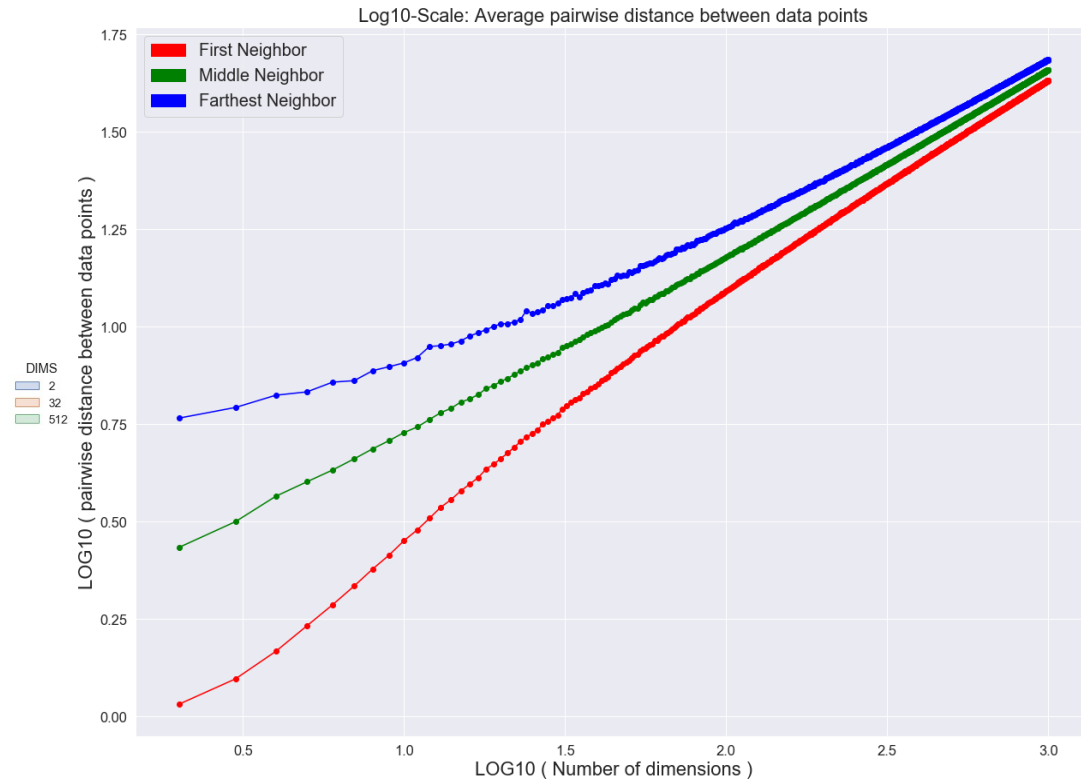
Ex.2 $E[\hat{\sigma}^2] = \frac{n-p}{n} \sigma^2$

Biased ML variance estimator in HD-space

Theoretical Background of the Curse of Dimensionality

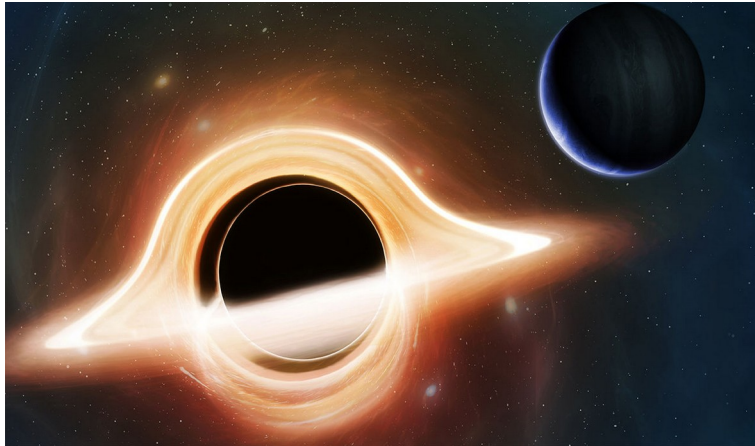
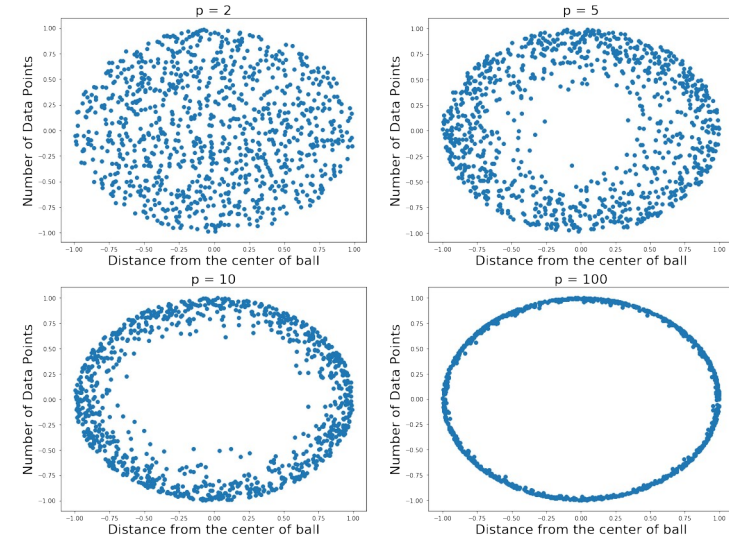
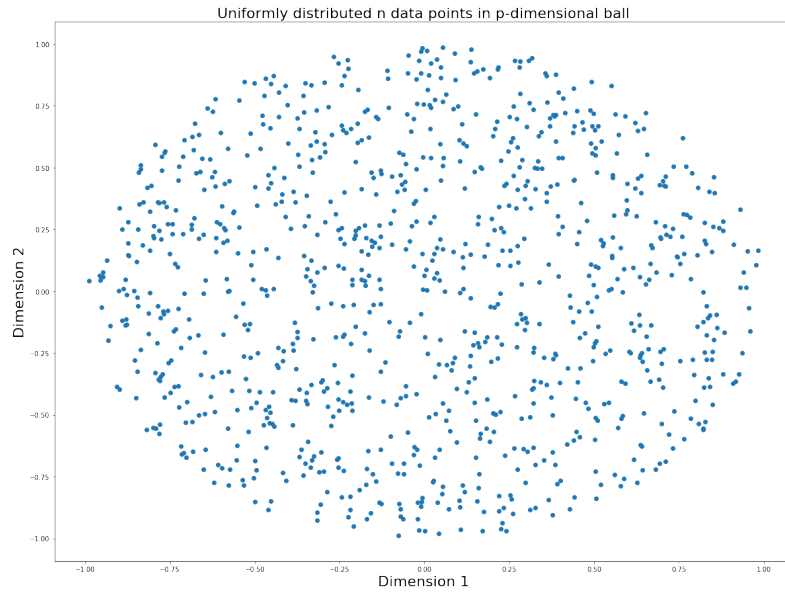


Data points become far from each other and equidistant in high dimensions



The differences between closest and farthest data point neighbours disappears in high-dimensional spaces: can't run cluster analysis

In high-dimensional space we can not separate cases and controls any more



High-dimensional data can be viewed as having a **“hole in the middle”**, hence the concept of mean / centroid loses its validity, hence we can't use Gaussian distribution

The curse(s) of dimensionality

There is such a thing as too much of a good thing.

Naomi Altman and Martin Krzywinski

We generally think that more information is better than less. However, in the 'big data' era, the sheer number of variables that can be collected from a single sample can be problematic. This embarrassment of riches is called the 'curse of dimensionality'¹ (CoD) and manifests itself in a variety of ways. This month, we discuss four important problems of dimensionality as it applies to data sparsity^{1,2}, multicollinearity³, multiple testing⁴ and overfitting⁵. These effects are amplified by poor data quality, which may increase with the number of variables.

Throughout, we use n to indicate the sample size from the population of interest and p to indicate the number of observed variables, some of which may have missing values for some samples. For example, we may have $n = 1,000$ subjects and $p = 200,000$ single-nucleotide polymorphisms (SNPs).

First, as the dimensionality p increases, the 'volume' that the samples may occupy grows rapidly. We can think of each of the n

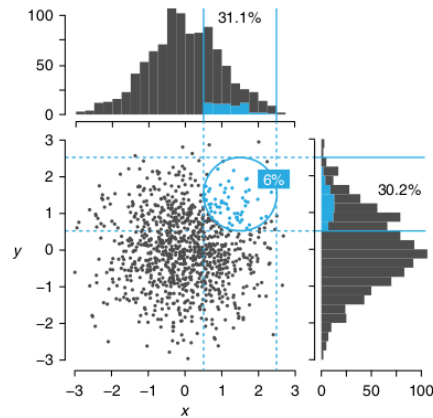


Fig. 1 | Data tend to be sparse in higher dimensions. Among 1,000 (x, y) points in which both x and y are normally distributed with a mean of 0 and s.d. $\sigma = 1$, only 6% fall within σ of $(x, y) = (1.5, 1.5)$ (blue circle). However, when the data are projected into a lower dimension—shown by histograms—about 30% of the points (all bins within blue solid lines) are within σ of 1.5. Blue

A and 100 to have the minor allele a . If we tabulate on two SNPs, A and B, we will expect only ten samples to exhibit both minor alleles with genotype ab . With SNPs A, B and C, we expect only one sample to have genotype abc , and with four or more SNPs, we expect empty cells in our table. We need a much larger sample size to observe samples with all the possible genotypes. As p increases, we may quickly find that there are no samples with similar values of a predictor.

Even with just five SNPs, our ability to predict and classify the samples is impeded because of the small number of subjects that have similar genotypes. In situations where there are many gene variants, this effect is exacerbated, and it may be very difficult to find affected subjects with similar genotypes and hence to predict or classify on the basis of genetic similarity.

If we treat the distance between points (e.g., Euclidian distance) as a measure of similarity, then we interpret greater distance as greater dissimilarity. As n increases, this

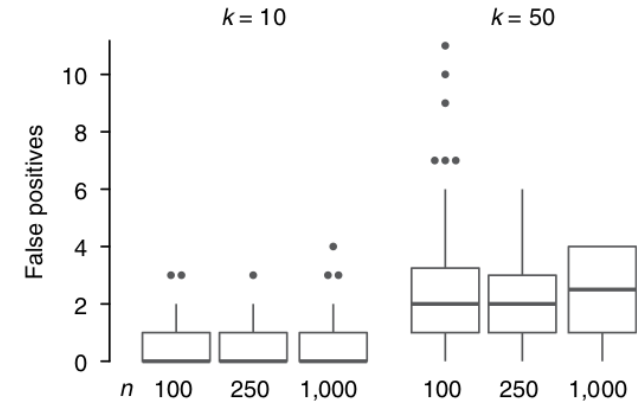


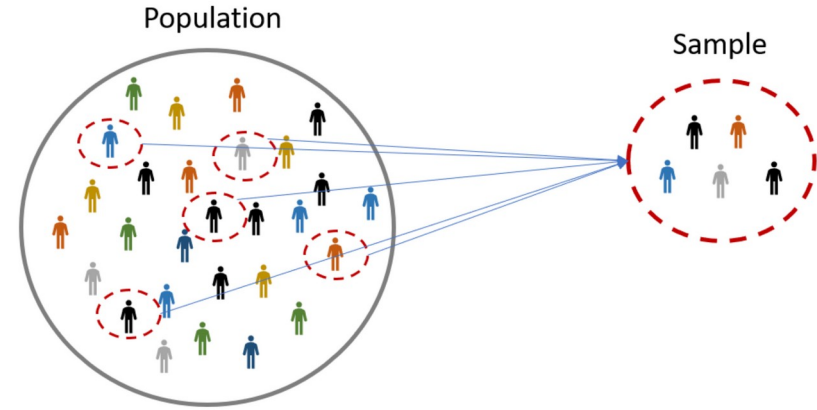
Fig. 3 | The number of false positives increases with each additional predictor. The box plots show the number of false positive regression-fit P values (tested at $\alpha = 0.05$) of 100 simulated multiple regression fits on various numbers of samples ($n = 100, 250$ and $1,000$) in the presence of one true predictor and $k = 10$ and 50 extraneous uncorrelated predictors. Box plots show means (black center lines), 25th and 75th percentiles (box edges), and minimum and maximum values (whiskers). Outliers (dots) are jittered.

Correcting for multiple testing does not solve the problem of too many false-positive hits



Regularization

Increase sample size: epidemiological limit

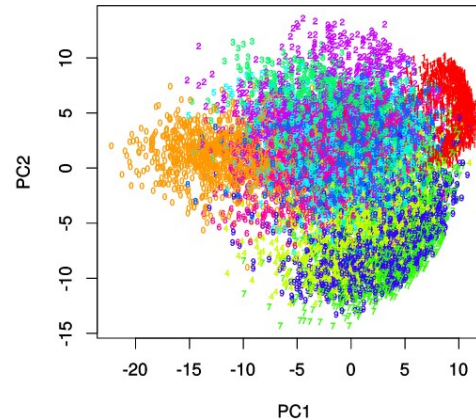


Dimension Reduction

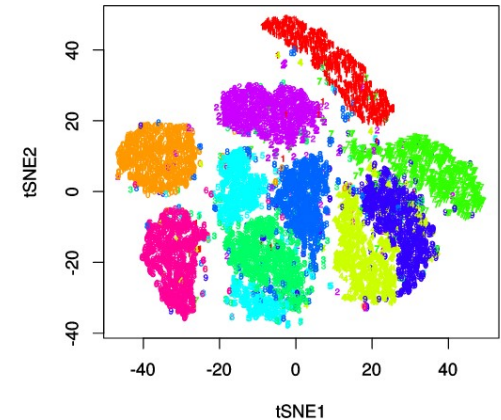


Bayesian statistics

PCA PLOT WITH PRCOMP



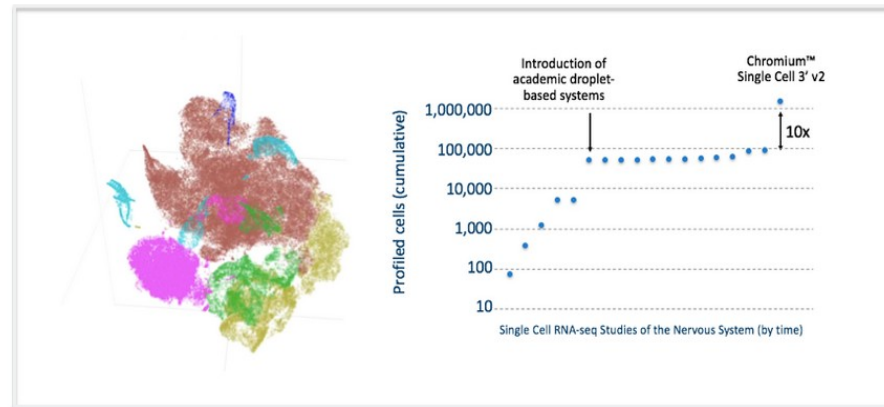
tSNE MNIST





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Our 1.3 million single cell dataset is ready to download



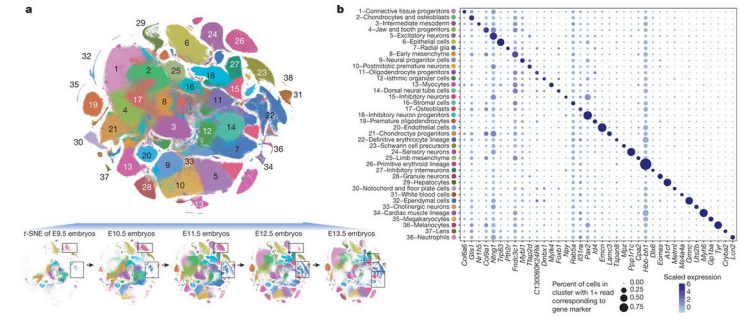
POSTED BY: **grace-10x**, on Feb 21, 2017 at 2:28 PM

At ASHG last year, we announced our 1.3 Million Brain Cell Dataset, which is, to date, the largest dataset published in the single cell RNA-sequencing (scRNA-seq) field. Using the Chromium™ Single Cell 3' Solution (v2 Chemistry), we were able to sequence and profile 1,308,421 individual cells from embryonic mice brains. Read more in our application note [Transcriptional Profiling of 1.3 Million Brain Cells with the Chromium™ Single Cell 3' Solution](#).



Fig. 2: Identifying the major cell types of mouse organogenesis.

From: *The single-cell transcriptional landscape of mammalian organogenesis*



a, t-SNE visualization of 2,026,641 mouse embryo cells (after removing a putative doublet cluster), coloured by cluster identity (ID) from Louvain clustering (in **b**), and annotated on the basis of marker genes. The same t-SNE is plotted below, showing only cells from each stage (cell numbers from left to right: $n = 151,000$ for E9.5; 370,279 for E10.5; 602,784 for E11.5; 468,088 for E12.5; 434,490 for E13.5). Primitive erythroid (transient) and definitive erythroid (expanding) clusters are boxed. **b**, Dot plot showing expression of one selected marker gene per cell type. The size of the dot encodes the percentage of cells within a cell type in

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Single Cell Analysis

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September

by biomembers • August 30, 2019

Human Cell Atlas, single-cell data

We are glad to announce that we will upsize the current single-cell database in [BioTuring Single-cell Browser](#) to 5,500,000 cells this September. With this release, we will double the current number of publications indexed in BioTuring Single-cell Browser, and cross the number of cells hosted on available public single-cell data repositories like [Human Cell Atlas \(HCA\)](#) and [Broad Institute's Single-cell Portal](#).

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RECENT POSTS

A new tool to interactively visualize single-cell objects (Seurat, Scanpy, SingleCellExperiments, ...)

September 26, 2019

5,500,000 cells will be indexed into BioTuring Single-cell Data Repository this September

August 30, 2019

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metabolomics
N $\approx$ P

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Proteomics

$N \approx P$

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Metagenomics

$N \approx P$

- manageable

$$\mathbf{N} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Transcriptomics
 $N \ll P$
 (Single cell: $N \leq P$)

challenging

[illegible]

Genomics
N <<<< P

[illegible]

Methylomics
N <<<< P

```
set.seed(123); n <- 20; p <- 2
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-2.0522	-0.6380	0.1451	0.3911	1.8829

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.14950	0.22949	0.651	0.523
X1	-0.09405	0.28245	-0.333	0.743
X2	-0.11919	0.24486	-0.487	0.633

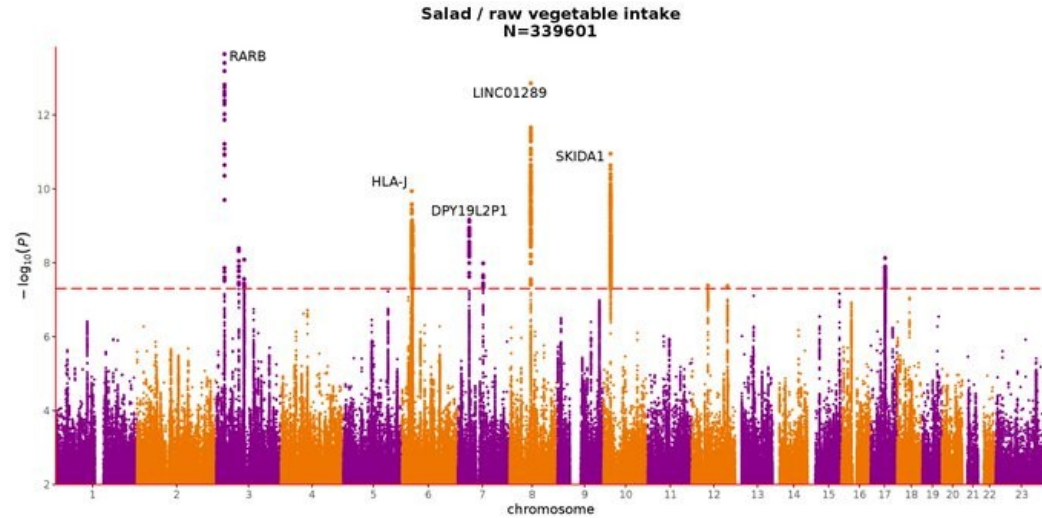
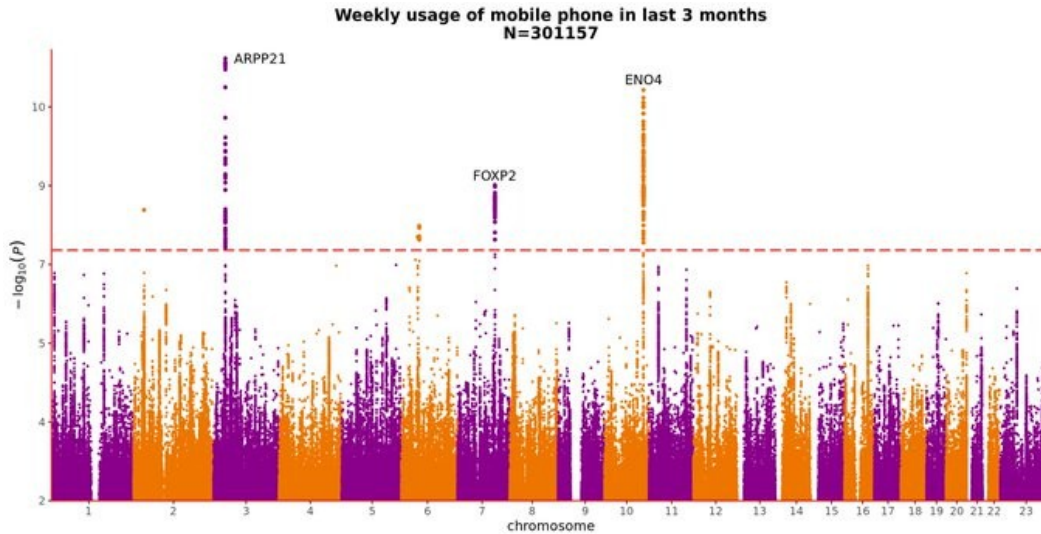
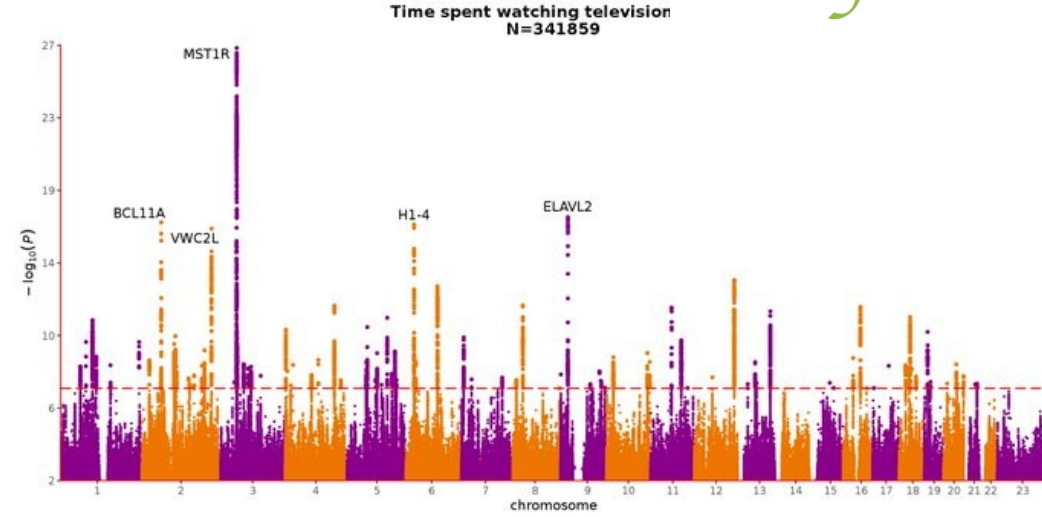
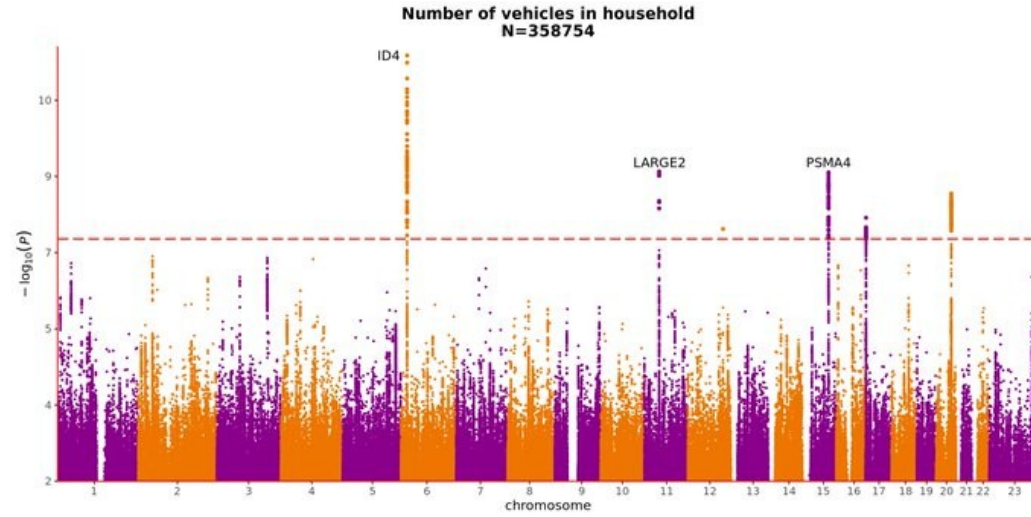
```
##
## Residual standard error: 1.017 on 17 degrees of freedom
## Multiple R-squared: 0.02204, Adjusted R-squared: -0.09301
## F-statistic: 0.1916 on 2 and 17 DF, p-value: 0.8274
```

```
set.seed(123456); n <- 20; p <- 10
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

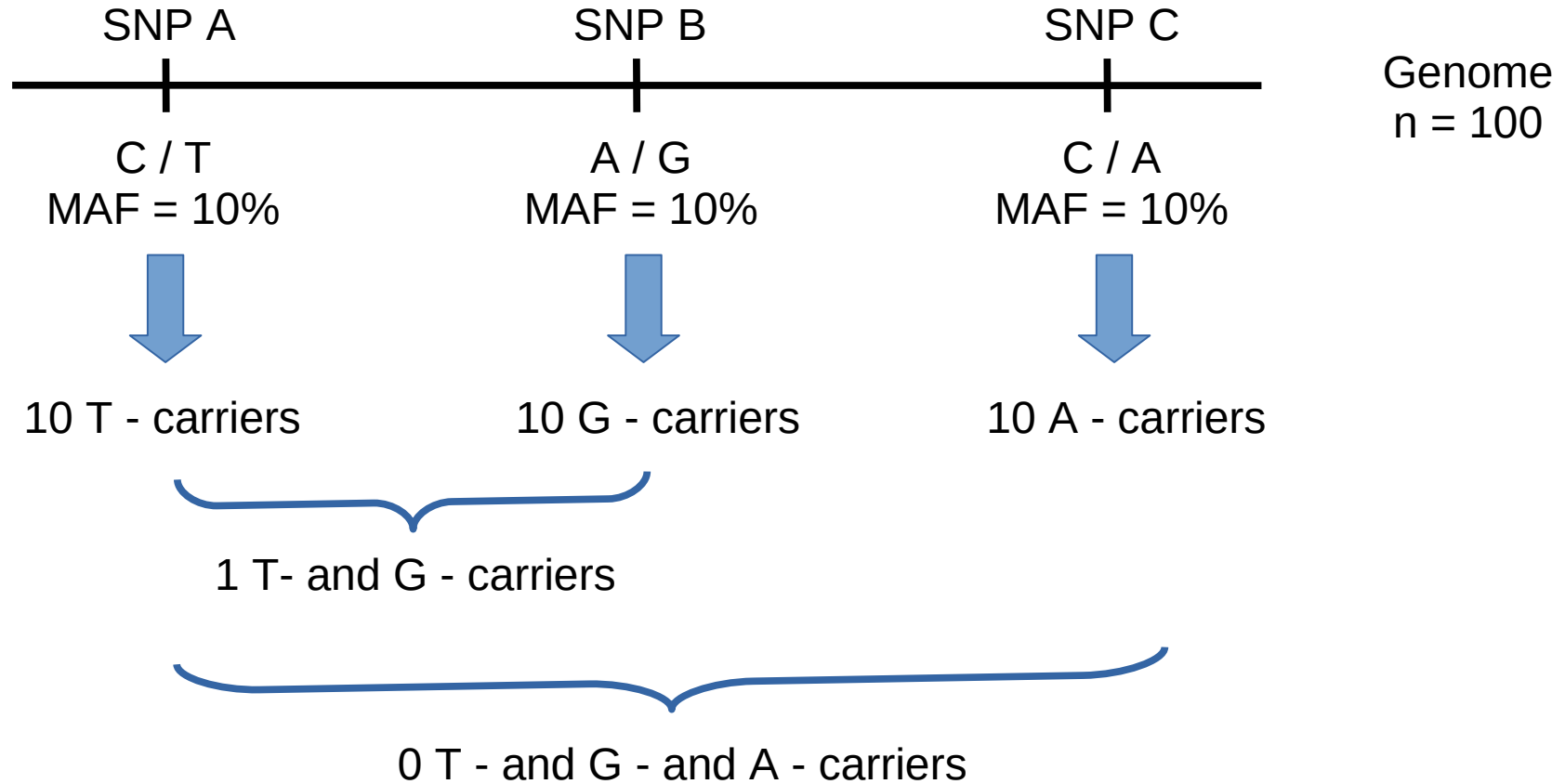
```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0255 -0.4320  0.1056  0.4493  1.0617
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.54916    0.26472   2.075   0.0679 .
## X1           0.30013    0.21690   1.384   0.1998
## X2           0.68053    0.27693   2.457   0.0363 *
## X3          -0.10675    0.26010  -0.410   0.6911
## X4          -0.21367    0.33690  -0.634   0.5417
## X5          -0.19123    0.31881  -0.600   0.5634
## X6           0.81074    0.25221   3.214   0.0106 *
## X7           0.09634    0.24143   0.399   0.6992
## X8          -0.29864    0.19004  -1.571   0.1505
## X9          -0.78175    0.35408  -2.208   0.0546 .
## X10          0.83736    0.36936   2.267   0.0496 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8692 on 9 degrees of freedom
## Multiple R-squared:  0.6592,    Adjusted R-squared:  0.2805
```

```
set.seed(123456); n <- 20; p <- 20
Y <- rnorm(n); X <- matrix(rnorm(n*p),n,p); summary(lm(Y~X))
```

```
##
## Call:
## lm(formula = Y ~ X)
##
## Residuals:
## ALL 20 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.34889      NA      NA      NA
## X1           0.66218      NA      NA      NA
## X2           0.76212      NA      NA      NA
## X3          -1.35033      NA      NA      NA
## X4          -0.57487      NA      NA      NA
## X5           0.02142      NA      NA      NA
## X6           0.40290      NA      NA      NA
## X7           0.03313      NA      NA      NA
## X8          -0.31983      NA      NA      NA
## X9          -0.92833      NA      NA      NA
## X10          0.18091      NA      NA      NA
## X11          -1.37618      NA      NA      NA
## X12          2.11438      NA      NA      NA
## X13          -1.75103      NA      NA      NA
## X14          -1.55073      NA      NA      NA
## X15          0.01112      NA      NA      NA
```



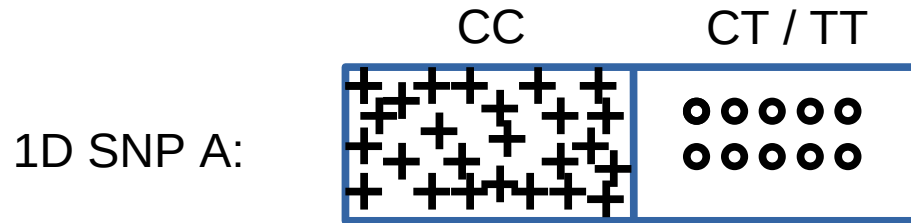
How The Curse of Dimensionality Affects Genomics Research



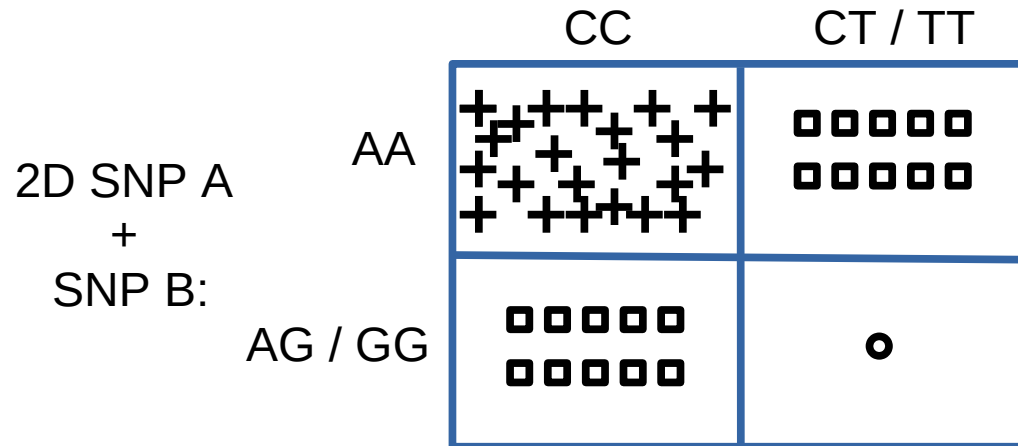
N = 100

SNP A (C / T, MAF = 10%) and SNP B (A / G, MAF = 10%)

Phen ~ SNP A + SNP B

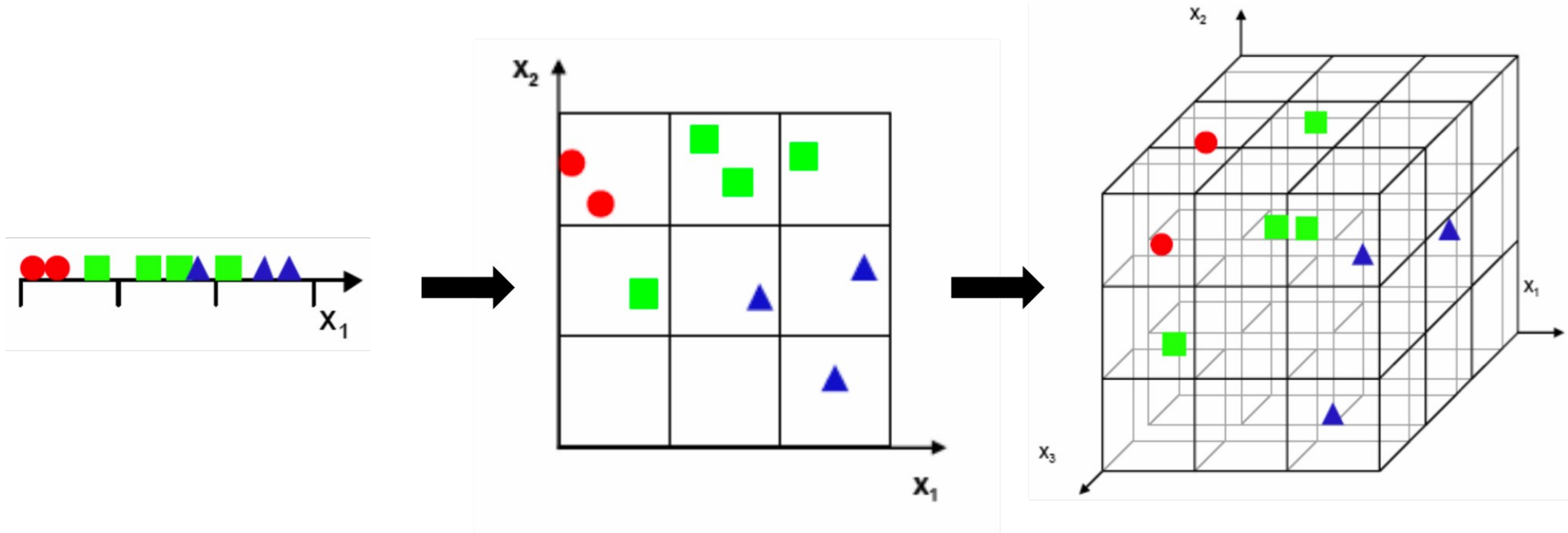


o - minor T – allele carrier



o - carrier of both minor T - and G - alleles

□ - carrier of either minor T - or G - alleles



Volume of high-dimensional data grows exponentially with the number of dimensions

A lot of free space appears, very difficult to sample (have observations for the whole range of values on each dimension / axis) that large space

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 National Cancer Institute
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TCGA's Pan-Cancer Atlas of insights and overarching themes on cancer culminates the 10+ year project.

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TCGA's Pan-Cancer Atlas

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Adult Soft Tissue Sarcomas

Cancers Selected for Study

TCGA in Action

December 2016
CASE STUDY: A Researcher Mined TCGA Data to Study Her Own Ovarian Cancer
 Faced the bleak prognosis associated with stage IIIC ovarian cancer, Shirley Pepke, Ph.D., used her computational biology background to study her own cancer in comparison to TCGA data. Pepke has now been in remission for a year. Image Credit: Gus Ruelas/USC

September 2016
CASE STUDY: TCGA Data Leveraged for Developing FDA-Designated Breakthrough Therapy
 Pharmaceutical company Loxo Oncology used TCGA data in the research that led them to develop FDA-designated Breakthrough Therapy LOXO-101, a promising new targeted therapy.

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June 12, 2018
Integrated Molecular Characterization of Testicular Germ Cell Tumors
 TCGA network researchers identify molecular characteristics that classify testicular germ cell tumor types, including a separate subset of seminomas defined by *KIT* mutations. This provides a set of candidate biomarkers for risk stratification and potential therapeutic targeting.

May 01, 2018
TCGA Releases The Pan-Cancer Atlas
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[Read more about Biobank UK](#)

New research links even low levels of air pollution with serious changes in the heart

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Video: "UK Biobank is what we always dreamed of"

GDPR information notice

Early-Career Researcher of the Year results!

Imaging: participants would recommend a friend

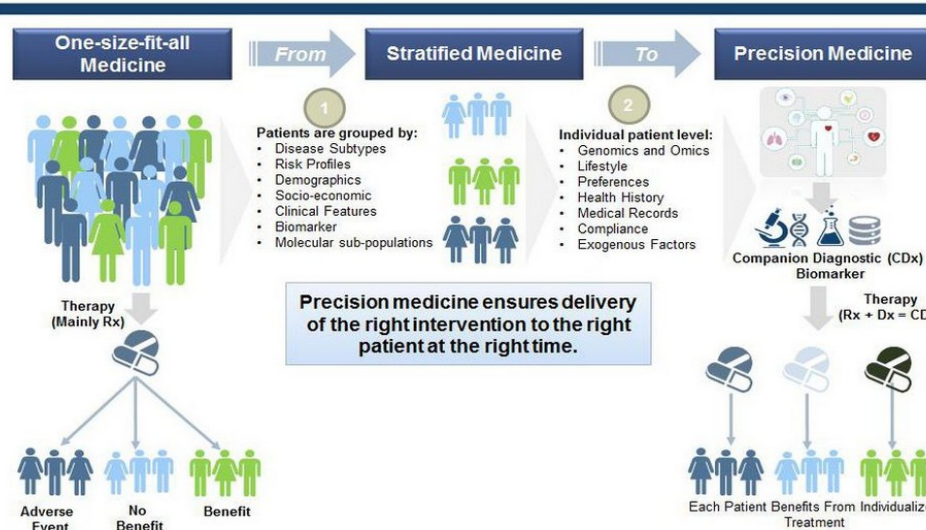


Mark McCarthy:
 Now we have found over 400 genetic signals influencing type 2 diabetes,
 what on earth are we going to do with them?

Precision Medicine: Why isn't it in the Clinics?

New Paradigm Shift in Treatment

Transitioning From the 'one-size-fits-all' to 'precision medicine' model with multi-level patient stratification.



Typical debate at Precision Medicine conferences:

- 1) Leaders: "We sequence a lot, it is important"
- 2) Clinicians: "This does not help us in the Clinics"

GRS = $\beta_1 \cdot \text{SNP1} + \beta_2 \cdot \text{SNP2} + \dots$ ← training

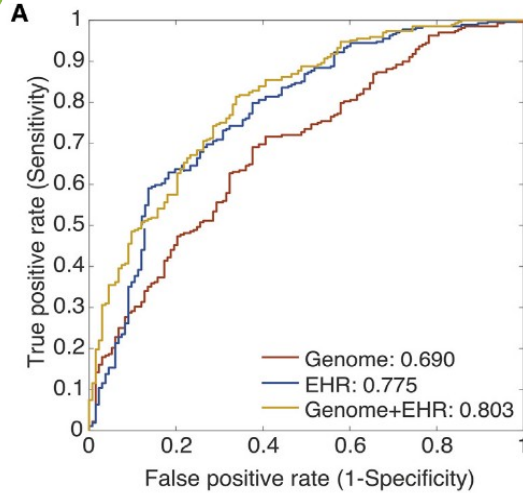
Phenotype $\sim$ GRS $\Rightarrow$ Adjusted R^2 ← validation

NEWS FEATURE PERSONAL GENOMES

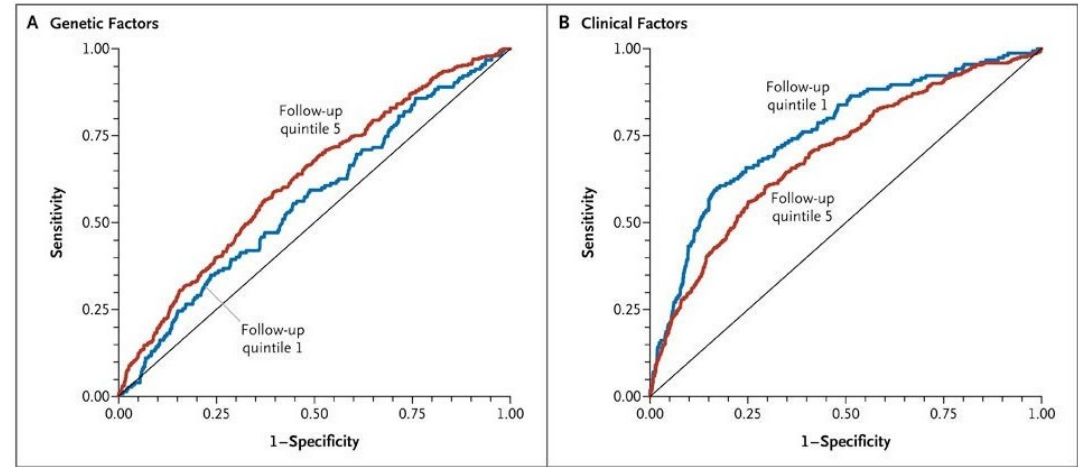
NATURE | Vol 456 | 6 November 2008



The case of the missing heritability



Li et al., 2018, Cell 174, 1361-1372
Decoding the Genomics of Abdominal Aortic Aneurism



Lyssenko et al., 2008, New England Journal of Medicine 359, 2220-2232
Clinical Risk Factors, DNA Variants and Development of Type 2 Diabetes



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bioRxiv is receiving many new papers on coronavirus 2019-nCoV. A reminder: these are preliminary reports that guide clinical practice/health-related behavior, or be reported in news media as established information.

New Results

2 comments

The predictive power of the microbiome exceeds that of genome-wide association studies in the discrimination of complex human disease

Braden T Tierney, Yixuan He, George M Church, Eran Segal, Aleksandar D Kostic, Chirag J Patel

doi: <https://doi.org/10.1101/2019.12.31.891978>

This article is a preprint and has not been certified by peer review [what does this mean?].

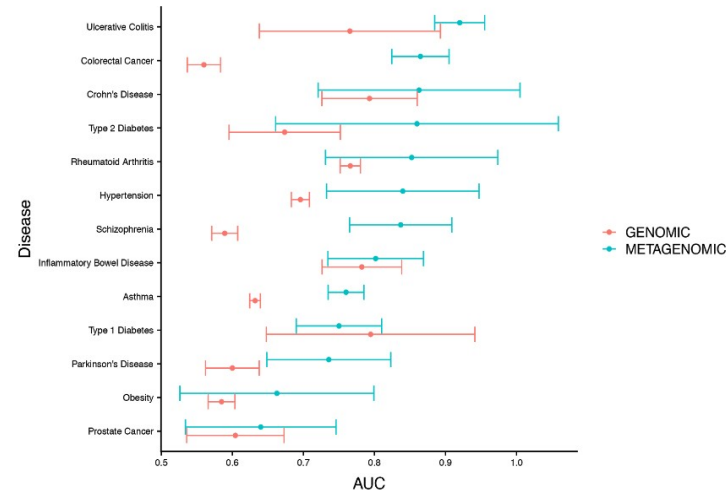
Abstract

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NEWS • 23 APRIL 2019

Genetic study homes in on height's heritability mystery

Analysis of 20,000 genomes reveals that rare gene variants can help to explain how the trait is inherited.

Linda Geddes



Height is often inherited — but the precise genetics factors have long eluded researchers. Credit: Shomos Uddin/Getty

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Link First 'tall gene' found

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bioRxiv is receiving many new papers on coronavirus SARS-CoV-2. A reminder: these are preliminary reports that have not been peer-reviewed. They should not be regarded as conclusive, guide clinical practice/health-related behavior, or be reported in news media as established information.

Recovery of trait heritability from whole genome sequence data

Pierrick Wainschein, Deepti P. Jain, Loic Yengo, Zhili Zheng, TOPMed Anthropometry Working Group, Trans-Omics for Precision Medicine Consortium, L. Adrienne Cupples, Aladdin H. Shadyab, Barbara McKnight, Benjamin M. Shoemaker, Braxton D. Mitchell, Bruce M. Psaty, Charles Kooperberg, Dan Roden, Dawood Darbar, Donna K. Arnett, Elizabeth A. Regan, Eric Boerwinkle, Jerome I. Rotter, Matthew A. Allison, Merry-Lynn N. McDonald, Mina K Chung, Nicholas L. Smith, Patrick T. Ellinor, Ramachandran S. Vasan, Rasika A. Mathias, Stephen S. Rich, Susan R. Heckbert, Susan Redline, Xiuqing Guo, Y.-D. Ida Chen, Ching-Ti Liu, Mariza de Andrade, Lisa R. Yanek, Christine M. Albert, Ryan D. Hernandez, Stephen T. McGarvey, Kari E. North, Leslie A. Lange, Bruce S. Weir, Cathy C. Laurie, Jian Yang, Peter M. Visscher

doi: <https://doi.org/10.1101/588020>

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Abstract

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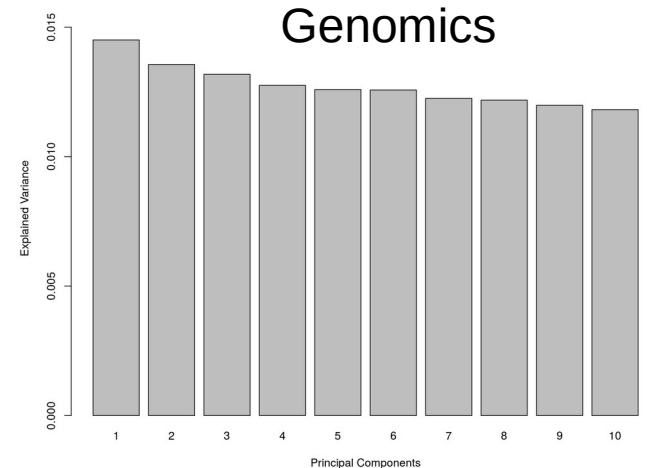
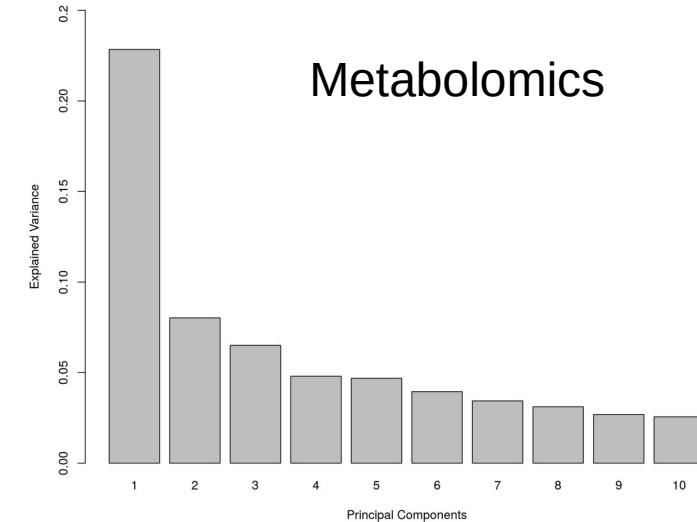
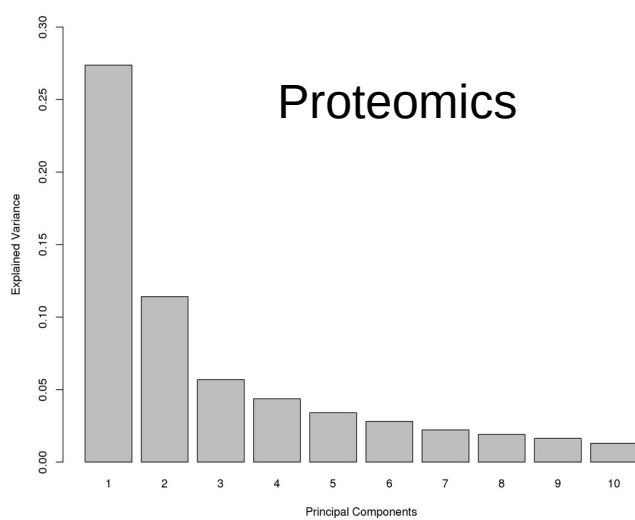
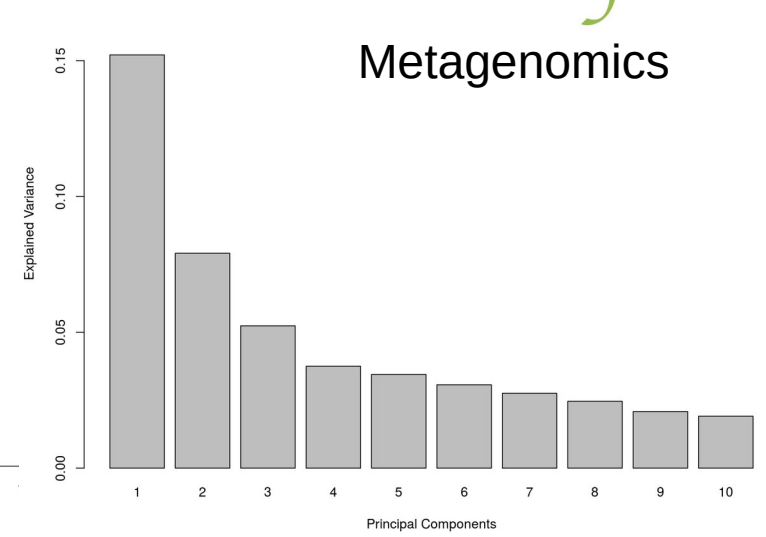
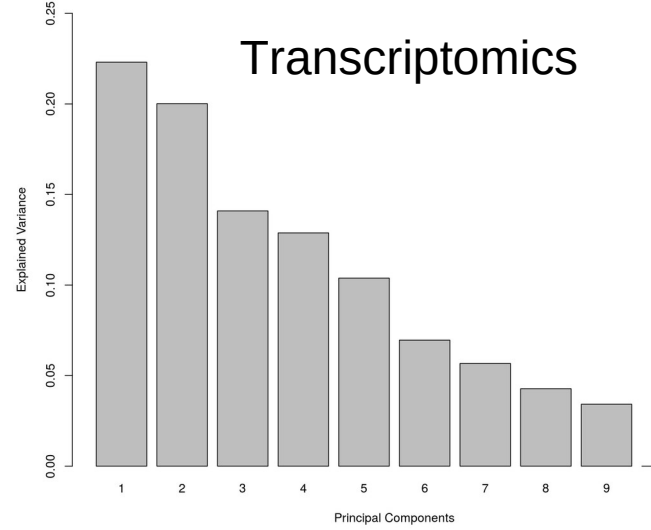
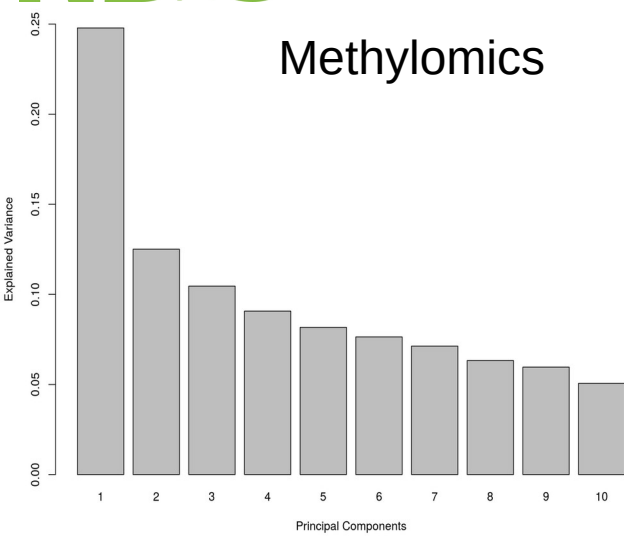
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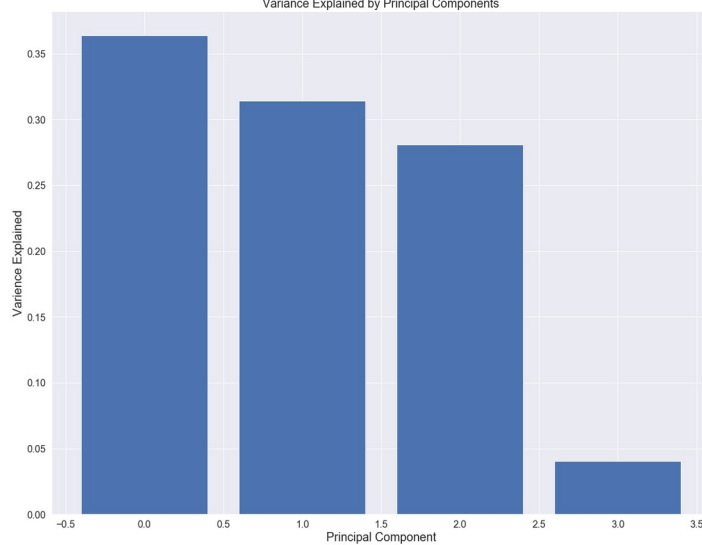
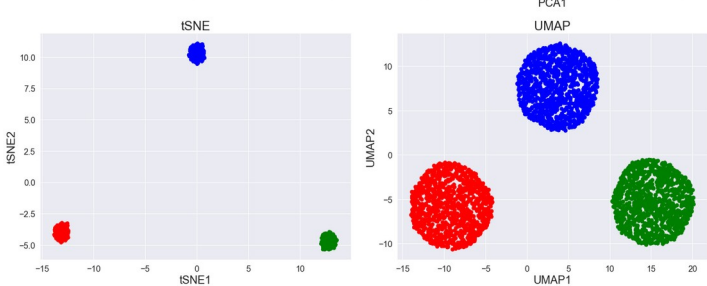
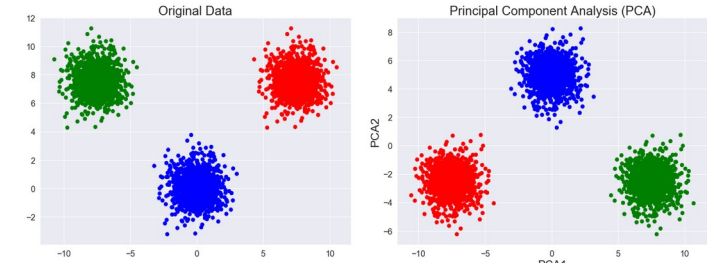
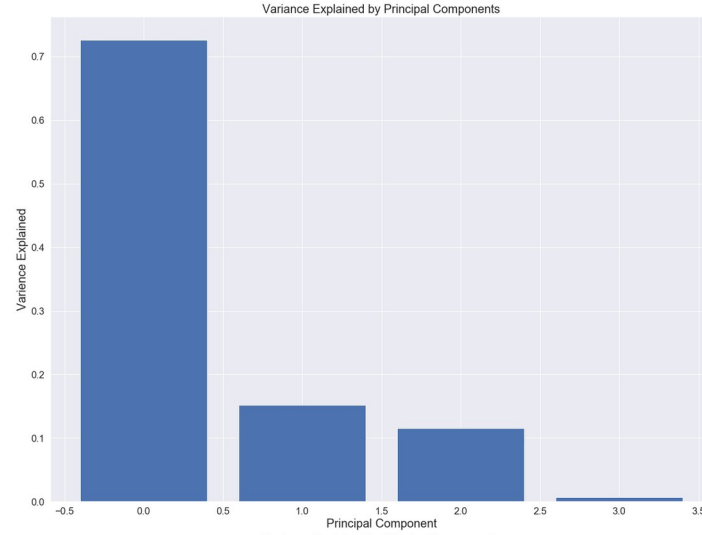
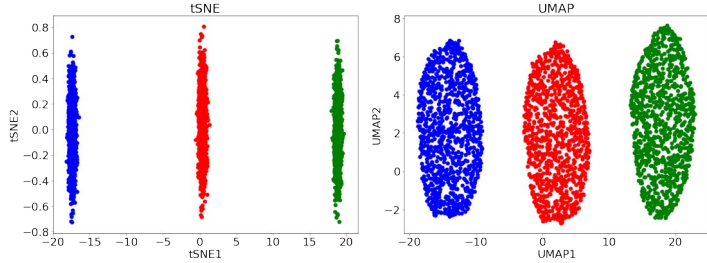
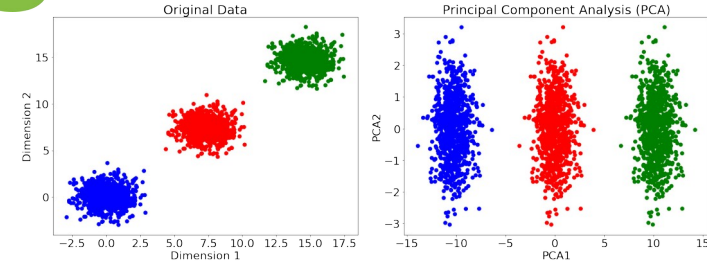
Metrics

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Abstract

Heritability, the proportion of phenotypic variance explained by genetic factors, can be estimated from pedigree data ¹, but such estimates are uninformative with respect to the underlying genetic architecture. Analyses of data from genome-wide association studies (GWAS) on unrelated individuals have shown that for human traits and disease, approximately one-third to two-thirds of heritability is captured by common SNPs ^{2–5}. It is not known whether the remaining heritability is due to the imperfect tagging of causal variants by common SNPs, in particular if the causal variants are rare, or other reasons such as over-estimation of heritability from pedigree data. Here we show that pedigree heritability for height and body mass index (BMI) appears to be fully recovered from whole-genome sequence (WGS) data on 21,620 unrelated individuals of European ancestry. We assigned 47.1 million genetic variants







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